

ISHAAN RANJAN

Applied AI & Software Engineer | ML Research · Agent Systems · Quantitative Modeling · Product
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SUMMARY

Builder and researcher creating applied AI systems for messy real-world decision problems, combining machine learning, statistics, forecasting, and usable software. Co-founder of an applied-AI studio with 5,000+ users and \$48K+ in non-dilutive funding, with published research spanning policy forecasting, agent systems, fine-tuned small models, and environmental engineering. Incoming Georgia Tech CS (Intelligence + Theory) on a 2.5-year track.

EDUCATION

Georgia Institute of Technology — Atlanta, GA Begins Fall 2026 · Expected Dec 2028

B.S. Computer Science, Threads: Intelligence and Theory · Accelerated 2.5-year track via AP and transfer credit

BASIS Peoria — Peoria, AZ

May 2026

High School Diploma · GPA 3.99 UW / 4.87 W · SAT 1540 · ACT 35 · 15 AP exams (14 scores of 5, one 4) · AP Scholar with Distinction ·

BASIS Educational Ventures Scholar (\$5,000) · #1 U.S. public high school (U.S. News 2024)

SKILLS

Programming: Python, Java, JavaScript/TypeScript, Swift, SQL, R, HTML/CSS

AI / ML: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, LLMs, LoRA & supervised fine-tuning, small language models, multi-agent systems, LLM evaluation, benchmark design, prompt engineering, Model Context Protocol (MCP), graph neural networks, NLP

Data / Quant / Research: pandas, NumPy, Jupyter, causal inference, difference-in-differences, regression, forecasting, experimental design, statistics, reproducible analysis, data visualization, technical writing

Product / Infra: FastAPI, React, Electron, Streamlit, Supabase, Vercel, REST APIs, Playwright, Git/GitHub, Arduino/IoT, on-device AI

Business / Communication: structured problem solving, market & user research, stakeholder interviews, product strategy, KPI dashboards, competitive analysis, public speaking

EXPERIENCE

Luxen LLC — Co-Founder

Dec 2023 – Present

Applied-AI studio for agent tooling, research dashboards, and workflow automation

- Led product strategy from prototype to public deployment across AI agents, developer tools, and research workflows; converted 100+ user interviews into roadmap and technical decisions.
- Raised \$48K+ in non-dilutive grants, startup support, and cloud credits through 1517 Fund, Microsoft for Startups, and BizWorld; earned 1517 Medici recognition and 2x Z Fellows Finalist distinction.
- Scaled products to 5,000+ active users and 1.5M+ media views while maintaining public, reproducible technical artifacts.

Links: luxenai.org · github.com/LuxenAI/agent-tree · github.com/LuxenAI/slmharness

Y Combinator Startup School 2026 — Incoming Founder

Summer 2026

Selected founder programming; startup credits and in-person access

- Selected for YC Startup School 2026; using customer discovery, competitive analysis, and rapid prototyping to pressure-test AI product strategy and go-to-market assumptions for Luxen/agent-tooling products.

University of Arizona, Precision Aging Network Lab — KEYS Research Intern

Summer 2025

Biostatistics, epigenetic aging, and translational research

- Analyzed vitamin D, omega-3, and exercise interventions against epigenetic aging clocks (Horvath, AltumAge, DunedinPACE) in R and Python with reproducible notebooks; reported and stress-tested a null result with attention to confounding.
- Presented a poster talk to 800+ attendees, translating biological-aging models into clear visuals for a mixed audience.

Links: [KEYS project site](#) · [abstract PDF](#) · [talk](#)

SELECTED PROJECTS

PRISM: Alcohol Policy Impact Atlas | Python, Causal Inference, NLP, Forecasting, Streamlit

- Built a 51-state, 2000–2022 policy panel with event-study difference-in-differences models, automating 15+ figures and a public decision-support dashboard.
- Designed mechanism-aware classification, causal-audit, and crash-risk forecasting modules: Macro-F1 0.962 on a 250-row audited benchmark; RandomForest RMSE 0.863 / R^2 0.454 on a held-out 2020–2023 window, with transparent assumptions and reproducible artifacts.

Links: github.com/IshaanAyaan/prism-policy-ai · [dashboard](#) · [SSRN DOI](#)

Specialized Small-Model Subagents + slmharness | PyTorch, LoRA, Fine-Tuning, MCP, Multi-Agent Systems

- Fine-tuned small language models into task-specific coding subagents; a 3B specialist matched a 72B baseline at far lower cost per successful task.
- Built 135M–1.5B specialists and slmharness, a local-first orchestration harness routing subtasks to cheap reliable executors before escalating to frontier models; paper in progress for an ML/agents workshop.

Links: github.com/LuxenAI/slmharness

AgentTree | JavaScript, Electron, AI Agents, Developer Tools

- Built an open-source, local-first workspace for hierarchical master/subagent coding sessions with shared context, live agent status, runtime model switching (Claude/Codex), and session control; 5,000+ users.

Links: github.com/LuxenAI/agent-tree

YConstruction | SwiftUI, Gemma 3n, Whisper, Qwen RAG, Supabase, IFC/BIM

- Built a voice-first iPhone app for construction defect reporting: on-device transcription, AI issue structuring, offline queueing, Supabase sync, and BIM-grounded architect review in Blender/Bonsai.
- Constrained model outputs to project-specific IFC vocabulary and emitted BCF topics; selected for a live pitch to YC investors at the YC x Google DeepMind hackathon.

Links: github.com/IshaanAyaan/YConstruction · [Devpost](#) · [demo](#)

PharmaFlow | Graph Neural Networks, Temporal Modeling

- Modeled pharmaceutical payments and prescribing as a temporal graph to flag influence patterns; prototype reached 82% accuracy on a dataset constructed to mirror CMS Open Payments distributions.

DebatICA | TypeScript, Next.js, Supabase, NLP, Web Crawling, APIs

- Shipped a debate research and round-prep platform used by 2,000+ debaters; built crawling, deduplication, and source-organization pipelines that cut research time while keeping argument synthesis with the user.

Links: debatICA.org · github.com/IshaanAyaan/debaticav1

S.A.B.E.R. Bioreactor | Arduino/IoT, Random Forest, Environmental Engineering

- Built an AI-automated *Chlorella vulgaris* bioreactor using pH, turbidity, and CO₂/O₂ sensors with pump-control logic to stabilize growth under variable conditions. AZSEF Grand Award; published preprint.

Links: [ResearchGate preprint](#)

RESEARCH & PUBLICATIONS

- PRISM: Alcohol Policy Impact Atlas — mechanism-aware policy forecasting and scenario simulation. [SSRN](#) · [dashboard](#)
- Criminal Tribes Act and Social Exclusion — colonial law and long-run social exclusion; ranked #1 in downloads, National History Day national qualifier. [SSRN](#)
- S.A.B.E.R.: Sustainable AI-automated Bioreactor for Environmental Remediation. [ResearchGate](#)
- KEYS @ Precision Aging Network — epigenetic aging analysis. [project site](#) · [abstract PDF](#) · [talk](#) · [Google Scholar](#)

LEADERSHIP & COMMUNITY

- Speech & Debate Captain (110+ member team) — Arizona State Champion; NSDA Big Questions Top 5 nationally; NSDA Nationals qualifier; built case files, drills, and evidence workflows for team-wide performance.
- Research Club President & Science Olympiad Engineering Lead — founded a student research program, mentored 30+ students through 10+ papers; Wind Power first place at the Gopher Invitational.
- AI Literacy Initiative — co-designed classroom tools and guides with teachers, reaching 150+ students and faculty while preserving student reasoning.
- MIT Blueprint — built GreenhouseAgent / A.C.R.E., an Arduino/FastAPI/OpenAI greenhouse-automation stack with live dashboard and Supabase logging. University of Austin Palantir Conference — selected attendee. [GitHub](#) · [Devpost](#)
- Student Council President; shipped Know Your Rights AZ, a civic resource hub for Phoenix-area residents distributed via QR codes at partner libraries.

AWARDS & DISTINCTIONS

- **Entrepreneurship:** 1517 Fund Medici Grantee; 2x Z Fellows Finalist (2024, 2025); BizWorld YES! Accelerator; Microsoft for Startups credits; YC Startup School 2026; \$48K+ in funding and credits; 5,000+ users; 1.5M+ media views.
- **Research & STEM:** Georgia Tech Stamps + Gold Scholarship Semifinalist; AZSEF Grand Award; SSRN paper ranked #1 in downloads; Science Olympiad Wind Power #1 (Gopher Invitational); National Economics Olympiad Top 43; U.S. Medicine Olympiad Honorable Mention.
- **Academics:** SAT 1540; ACT 35; 15 AP exams (14 scores of 5, one 4); AP Scholar with Distinction; GPA 3.99 UW / 4.87 W; BASIS Educational Ventures Scholar (\$5,000).